

and sacrum bones on each side of the pelvis.

SI-BONE has developed a proprietary 3D printing technology in order to develop the implant. The implant features an enhanced porous surface that resembles the trabecular structure of cancellous bone and a unique fenestrated design; both of these features combine to make an environment that promotes bone ongrowth, ingrowth, through growth and intra-articular fusion. The fenestrated structure and surface were specifically designed to promote intra-articular fusion and offer enhanced osteointegration.

Similarly, Additive Orthopaedics has received FDA clearance for its 3D printed minimally-invasive, intramedullary Bunion Correction System that helps to address alignment and stabilisation of bunions. According to a doctor, this 3D printed implant enables certain features in plates, which cannot be achieved through standard manufacturing processes. For example, one area of the plate can be made to provide additional fixation and strength, to enable the bone to grow onto and through the plate over time. This will support better integration and alignment of bones following a bunion procedure.

Johnson & Johnson, one of the popular names in healthcare, has tied up with companies like Organovo and HP to evaluate the use of 3D bio-printed tissues and other 3D printed healthcare solutions. They have also collaborated with Inkbit, a start-up spun out of the Massachusetts Institute of Technology (MIT), which has come up with a unique multi-material 3D printing technology to print complex consumer products with integrated electronics. In April this year, the group also acquired the 3D printing technology from Tissue Regeneration Systems Inc (TRS), to create patient-specific, bioresorbable implants with a unique

mineral coating. These implants can be used to support bone healing in patients with orthopaedic and craniomaxillofacial deformities and injuries.

In June this year, a team of researchers at the Boston University College of Engineering revealed a potential treatment for ischemia using a cell-infused 3D printed patch that can guide the growth of new blood vessels. The team tied up with a Boston-based biomedical tech company called Innolign to 3D

The Boeing 787 Dreamliner will be the first commercial airplane to fly with a certified, structural, 3D printed titanium part!

print vessels on a small scale of 100 microns. The cells had to be infused into the patches in a particular pattern in order to be effective, and that is where 3D printing proved its worth. The inherent flexibility of 3D printing allowed the team to change and test their designs, helping them to discern which patterns worked best. 3D printing also offers them the scalability needed to test their designs in larger, more complex organisms and tissue environments in the future.

From shoes to electronics, everything can be 3D printed

A four-year-old company called Carbon has come up with a photochemical process that uses light and oxygen to produce objects from pools of liquid polymer. This enables faster printing, greater structural integrity and better consistency than conventional methods. The technology is based on stereolithography (SL).

Objects are constructed layer by layer by shining a light into a pool



SINCE 1962

INSULATION TESTERS

Analog



MC-9008A Series

Model	Range	Test Voltage DC
MC-9008A	0-20 M Ohms	300V
MC-9008A	0-100 M Ohms	300V
MC-9008A	0-500 M Ohms	300V
MC-9008A	0-1000 M Ohms	300V
MC-9008A	0-200 M Ohms	1000V
MC-9008A	0-500 M Ohms	1000V
MC-9008A	0-2000 M Ohms	1000V

Digital



DIT 99 Series

Model	Range	Test Voltage DC	Resolution
DIT 99A	0-20 M Ohms	300V	0.01 M Ohms
DIT 99B	0-200 M Ohms	250V	0.1 M Ohms
DIT 99C	0-200 M Ohms	500V	0.1 M Ohms
DIT 99D	0-200 M Ohms	1000V	0.1 M Ohms
DIT 99E	0-2000 M Ohms	1000V	1 M Ohms

5KV



DIT 95A

Specification	Test Voltage	Range
Insulation Resistance	1000V / 2500V / 5000V	0.1M to 2000G
AC Voltage Measurement	0-600VAC (50-60Hz)	-
Phase Sequence Test	100V-400V (Phase - Phase) 40-60Hz	>

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